

December 2018

MANUFACTURING PROCESSES – I*Time Allowed: 3 Hours**Full Marks: 70***Answer to Question No.1 is compulsory and to be answered first.****This answer is to be made in separate loose script(s) provided for the purpose.****Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.****On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.****Answer any five questions from the rest.**

- I. Answer the following questions (any twenty): 1x20
- i) Mechanical property desired for successful forging is – (a) Ductility, (b) Malleability, (c) Hardness, (d) Toughness
 - ii) Seamless tubes in mass production are manufactured by – (a) Drawing, (b) Extrusion, (c) Rolling, (d) Casting.
 - iii) In drop forging, forging is done by dropping – (a) The work piece at high velocity, (b) The hammer at high velocity, (c) The die with hammer with high velocity, (d) A weight on hammer to produce the requisite impact.
 - iv) Blanking and punching operations can be performed simultaneously on – (a) Combination die, (b) compound die, (c) progressive die, (d) simple die.
 - v) Fluxes are used while welding – (a) To increase the rate of welding, (b) To clean the joint, (c) To prevent oxidation of metal during welding, (d) All the above.
 - vi) Oxy-acetylene flame is used for welding of – (a) Monel, (b) Nickel, (c) Steel, (d) Bronze.
 - vii) The hidden defect of a weld can be detected by – (a) Tensile test, (b) Bend test, (c) Radiography, (d) All.
 - viii) Commonly used gases in TIG welding are – (a) Hydrogen & Oxygen, (b) Hydrogen & Argon, (c) Hydrogen & Helium, (d) Helium & Argon.
 - ix) Consumable electrode is used in the following process – (a) TIG welding, (b) Gas welding, (c) Arc welding, (d) Resistance welding.
 - x) Above _____ temperature, working of metals is named hot working.
 - xi) Drawing is done to _____ the diameter of wire.
 - xii) In 4-high rolling mill, bigger rolls are known as _____ rolls.
 - xiii) Accurate taper and step tapers can be machined with _____ attachment.
 - xiv) Finishing and sizing a hole is called _____.
 - xv) A twist drill has _____ flutes.

- xvi) Pattern used in precision investment casting normally made of _____.
- xvii) In resistance welding _____ is generated by the resistance offered to the flow of current.
- xviii) MIG welding uses _____ electrode.
- xix) In neutral flame in Oxyacetylene gas welding, ratio of acetylene to oxygen is _____.
- xx) Spot welding is one of the _____ welding processes.
- xxi) The revolving cutter of a milling machine is held on a _____.
- xxii) In universal milling machine table can have the _____ arrangement about a vertical axis.
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2. a) What is Pattern? Name different types of pattern. Sketch any three of them and explain.
 b) What are the pattern making allowances? Explain any one of them. 7+3
3. a) Discuss with sketches hot rolling and cold rolling.
 b) Discuss with neat sketches direct extrusion and indirect extrusion. 5+5
4. a) How do you specify a Lathe machine?
 b) Write short notes on orthogonal cutting and Oblique cutting.
 c) What are the different operations that can be done on a lathe machine? 3+5+2
5. a) How do you specify a Drilling machine? Draw a pillar type Drilling machine & mention basic parts.
 b) Write short notes on Up Milling & down Milling. 7+3
6. a) What is core? What is the purpose of core?
 b) Explain the defects in casting. How can those defects be prevented? 5+5
7. a) Explain Arc welding process showing all salient points on it.
 b) State various welding defects & explain the remedies. 5+5
8. a) Explain spot welding and show its simple set up.
 b) Name the different flames and show the different zones in each of them. 5+5
9. Write short notes on the followings (any four): 2½x4
 (a) Knurling, (b) Reaming, (c) Brazing, (d) Soldering, (e) Core Prints, (f) Centrifugal Casting, (g) Die Casting
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